BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: MRD-IX-21(A) / MRD-IX (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-IX-21(A)

Note: Note: There are three sections in this paper i.e., A, B & C. Instructions: Please attempt section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, overwriting or multiple circle shading.

Q1. Write the correct option i.e. A, B, C or D in the empty box provided opposite to each part. Cutting, erasing and over writing will not be awarded any marks.

Q1. Which of the following is a derived physical quantity.

(A) time
(B) electric current
(C) mass
(D) force

Q2. A body accelerates uniformly from $15ms^{-1}$ to $20ms^{-1}$ in 5 s. What is the acceleration of the body.

(A) $2ms^{-2}$
(B) $1.5ms^{-2}$
(C) $1ms^{-2}$
(D) none of these

Q3. Force needed to produce an acceleration of $10ms^{-2}$ in a ball of mass 0.5kg is.....

(A) 20N
(B) 10.5N
(C) 9.5N
(D) 5N

Q4. Which of the following forces can act as a centripetal force?

(A) tension
(B) friction
(C) gravitational force
(D) all of these

Q5. Conventionally the direction of anticlockwise torque is taken as:

(A) positive
(B) negative
(C) upward
(D) downward

Q6. The angle at which x and y components of force are equal is.....

(A) 0° (B) 30°

(C) 45° (D) 60°

Q7. The S.I unit for gravitational constant "G" is .....

(A) Nkg^2 (B) Nm^2Kg^{-2}

(C) Nm^2Kg^2 (D) $Nm^{-2}Kg^2$

Q8. A machine is able to lift 200N of concrete slab vertically up to a height of 30m above the ground in 50s. The average power of the machine is.....

(A) 1.33W (B) 60W

(C) 120W (D) 6000W

Q9. Young modulus is measured in units of .....

(A) kgm^{-3} (B) Pa

(C) N (D) None of these

Q10. A rock weighs 25.7N in air and 21.8N in water. What is the buoyant force of the water.

(A) 4.1 N (B) 3.9 N

(C) 1.18 N (D) 0.84 N

Q11. 310K temperature in centigrade scale is .....

(A) 37°C (B) 310°C

(C) 63°C (D) 273°C

Q12. The mode of transfer of heat which does not require material medium is called.....

(A) [illegible in source scan] (B) [illegible in source scan]

(C) [illegible in source scan] (D) [illegible in source scan]

Section - B

Maximum Marks: 32

QII. Answer any EIGHT parts. Each part carries FOUR marks. Note: Time allowed for Section - B and Section - C is 2 Hours and 45 minutes.

1. Why area is derived quantity?
2. Describe a situation in which the speed of an object is constant while the velocity is not.
3. How does friction help you walk? Is it kinetic friction or static friction?
4. Why does dust fly off, when a hanging carpet is beaten with a stick?

5. Explain why door handles are not put near hinges.

6. Moon is attracted by the earth, why it does not fall on earth?

7. Define energy and explain it with an example.

8. State Hooke's law. Write it mathematically.

9. Differentiate between heat and temperature.

10. The temperature of a normal human body is 37°C. Find this temperature on the Fahrenheit and Kelvin Scale?

11. Why white clothes are proffered wearing in summer? Explain briefly.

Section "C"

Maximum Marks: 21

> Note: Answer any THREE questions. All questions carry equal marks.

Q3.

a. What is momentum? Relate force to change in momentum.

b. Graphically prove that $S = vt + 1/2\ at^2$.

Q4.

a. What are rectangular components of a vector. Resolve a force F making an angle φ with X axis into its rectangular components.

b. A bolt on a car engine needs to be tightened with a torque of 40Nm. You use a 25-cm long wrench and pull on the end of the wrench perpendicularly. How much force do you have to exert?

Q5.

a. State and explain Newton's law of Gravitation.

b. At which altitude above Earth's surface would the gravitational acceleration be 4.9 m/s^2?

Q6.

a. State and explain Archimedes principle.

b. What is the specific heat of a metal substance if 135KJ of heat is needed to raise 4.1 kg of the metal from 18°C to 37.2°C.